TEXORA · LECTURE SHEET

The Drawing Frame

Drawing doubles several slivers and drafts them back to one, averaging out weight variation and straightening fibres. This sheet works the draft system (break draft and main draft), delivered sliver weight, the statistical effect of doubling, and production from delivery speed.

YM-3207 · ADVANCED SPINNING CALCULATIONS · TOPIC 4 · SPECIALIZATION — YARN MANUFACTURING

1. Objects of drawing

Card (or comber) sliver varies in weight from metre to metre and its fibres are hooked and only partly parallel. The drawframe attacks both problems with two mechanisms. **Doubling** feeds several slivers together — thick places in one are averaged against thin places in others. **Drafting** attenuates the combined strand between roller pairs running at increasing surface speeds, which pulls fibres straighter and reduces the delivered weight per metre back to roughly that of a single feed sliver. Two passages are normal (breaker and finisher), often with the finisher fitted with an autoleveller that measures the feed or delivery and trims the draft continuously.

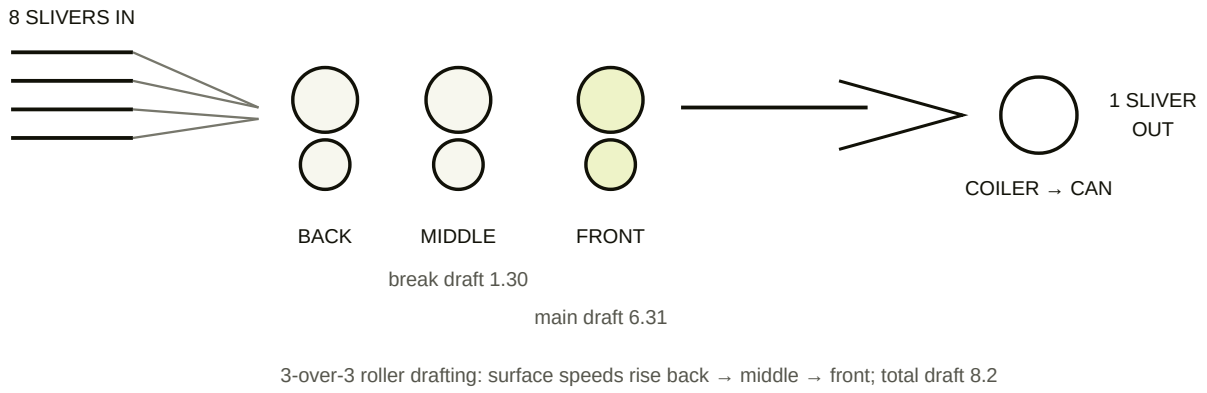


Figure 1 — Doubling and drafting: eight slivers enter, pass through three roller pairs with rising surface speeds, and leave as one sliver. Break draft acts between back and middle rollers, main draft between middle and front.

2. The draft system

The total draft is split between two zones so that fibres are first tensioned gently (break draft, just enough to straighten the strand and remove crimp) and then attenuated under control in the main zone.

$$\text{Total draft} = \text{break draft} \times \text{main draft}$$

WORKED EXAMPLE — DELIVERED SLIVER WEIGHT AND DRAFT DISTRIBUTION

Feed: 8 slivers of 3.94 g/m. Total draft = 8.2, split as break draft 1.30 and main draft to be found.
Combined feed weight = 8 × 3.94 = 31.52 g/m.
Delivered sliver = 31.52 / 8.2 = **3.84 g/m** (= 3841 tex, Ne 0.154).
Main draft = 8.2 / 1.30 = **6.31**.
Note the deliberate design: draft ≈ number of doublings, so the delivered sliver (3.84 g/m) is close to a single feed sliver (3.94 g/m). The drawframe's job is evenness and parallelisation, not large weight change.

Roller settings follow fibre length: the main-zone setting is set slightly longer than the mean fibre length so fibres are gripped by one roller pair at a time; too close a setting breaks fibres, too wide lets them float uncontrolled and creates drafting waves. In a 3-over-3 arrangement the middle top roller is often driven through an apron or kept slightly offset to improve control of short fibres.

3. What doubling does to evenness

If the feed slivers vary independently with coefficient of variation CV_{in}, the combined strand before drafting varies as the average of n independent strands:

$$\text{CV after doubling} = CV_{in} / \sqrt{n}$$

WORKED EXAMPLE — THE AVERAGING EFFECT

Feed sliver weight CV = 3.2%; doublings n = 8.
CV of combined strand = 3.2 / √8 = 3.2 / 2.83 = **1.13%**.
Drafting itself adds irregularity (floating fibres are never perfectly controlled), so the delivered sliver does not reach 1.13% in practice — but the √n law shows why doubling is such a powerful first step, and why two passages of 8 doublings (8 × 8 = 64 effective doublings) transform the product. The law assumes the slivers are independent; periodic faults that are in phase in all cans are not averaged away.

4. Production

$$\text{Production (kg/h)} = \text{delivery speed (m/min)} \times \text{delivered sliver (g/m)} \times 60 / 1000$$

WORKED EXAMPLE — DRAWFRAME PRODUCTION

Delivery speed = 450 m/min; delivered sliver = 3.84 g/m.
Production = 450 × 3.84 × 60 / 1000 = **103.7 kg/h** per delivery at 100%.
At 85% efficiency: **88.1 kg/h**.
A delivery therefore processes the output of roughly six to seven cards (13.0 kg/h each) — the arithmetic that sizes drawframe numbers in a spin plan.

5. Autolevelling

An autoleveller measures the incoming sliver thickness (or the delivered sliver) and continuously adjusts the draft so the delivered weight stays on target. It corrects medium- and short-term variation within its response length; it cannot remove a periodic fault and it cannot fix a feed that is wrong on average — that still requires a draft change wheel or its electronic equivalent. Control of the average (the A% test) is treated in YM-4101, Topic 5.

Glossary

Doubling — feeding several slivers together so their variations average out. **Draft** — the ratio of feed to delivered weight per unit length (equivalently, delivered to fed surface speed). **Break draft** — the small draft in the back zone that tensions the strand. **Main draft** — the principal attenuating draft in the front zone. **Floating fibre** — a fibre gripped by neither roller pair, moving uncontrolled. **Autoleveller** — a device that trims draft automatically to hold delivered weight constant. **Breaker / finisher** — first and second drawframe passages.

Key points

• Total draft = break draft × main draft (8.2 = 1.30 × 6.31 in the example).
• Delivered sliver = (doublings × feed weight)/total draft = 3.84 g/m in the example.
• Doubling cuts weight CV by √n — 3.2% → 1.13% for 8 doublings, before drafting adds its own irregularity.
• Production = delivery speed × sliver weight; 103.7 kg/h at 100%, 88.1 kg/h at 85%.
• Roller settings follow fibre length; autolevellers trim variation but not a wrong average.

Practice

1. Six slivers of 4.1 g/m are doubled with a total draft of 7.5. Calculate the delivered sliver weight in g/m and in tex.
2. A drawframe delivers at 500 m/min a sliver of 3.6 g/m. Calculate production at 100% and at 88% efficiency.

3. Feed sliver CV is 4.5%. What CV does the √n law predict after 6 doublings, and why is the real delivered CV higher?

Answer key

1. Combined feed = 6 × 4.1 = 24.6 g/m. Delivered = 24.6 / 7.5 = **3.28 g/m = 3280 tex**.
2. Production = 500 × 3.6 × 60/1000 = **108 kg/h**; at 88% = **95.0 kg/h**.
3. CV = 4.5 / √6 = 4.5 / 2.45 = **1.84%**. In practice drafting adds irregularity because floating fibres between roller pairs move at uncontrolled speeds, and roller eccentricity adds periodic variation, so delivered CV is higher than the pure doubling prediction.

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